30 September 1969
SUPERSEDING
MIL-N-43186
30 December 1963

MILITARY SPECIFICATION

NEEDLES, ETCHING, HAND

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope.- This specification covers hand, etching needles (see 6.1).

* 1.2 Classification.- The etching needles shall be of the following styles and sizes, as specified (see 6.2):

Style I - Round needle

Size 1 - 0.0312 inch diameter
Size 2 - 0.0625 inch diameter
Size 4 - 0.4687 inch diameter

Style II - Square needle

Size 1 - 0.027 inch square
Size 2 - 0.035 inch square
Size 3 - 0.040 inch square
Size 4 - 0.045 inch square
Size 5 - 0.055 inch square
Size 6 - 0.062 inch square
Size 7 - 0.075 inch square

Style III - Oval needle

Size 1 - 0.0312 inch by 0.0625 inch

Style IV - Flat needle

Size 1 - 0.0312 inch by 0.1875 inch

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 5120

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SPECIFICATIONS

FEDERAL

QQ-T-580 - Tool Steel, Carbon and Carbon Vanadium

MILITARY

MIL-H-15424 - Hand Tools; Packaging of

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publication.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E18-65 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article.- When specified (see 6.2), before production is commenced, a finished etching needle, covered by this specification, shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. Approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

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3.2 Materials.- Materials not definitely specified shall be of the quality normally used by the manufacturer of etching needles, provided the completed item complies with all the provisions of this specification.

* 3.2.1 Tool steel, carbon.- The tool steel shall conform to grade B, any W1 class, drill rod form, spheroidize annealed condition of QQ-T-580.

3.2.2 Wood.- The wood shall be any one of the following species: selected birch, beech, maple or hard wood dowel stock.

3.3 Design and construction.-

* 3.3.1 Needles, etching.- The etching needles shall conform to the design, details, and dimensions shown on figure 1 and to the requirements specified herein. The etching needle shall consist of a stylus like tool; a needle centered in a wooden handle.

* 3.3.1.1 Needle.- The needle shall be fabricated of the material specified in 3.2.1 and shall have a Rockwell hardness of C60-67 when tested as specified in 4.4.1. The needle shall be centered, set, and secured into the handle specified in 3.3.1.2. The method used in securing the needle into the handle shall conform to the method normally used by the manufacturer in securing commercial etching needles. The needle shall be ground to the dimensions shown on figure 1.

* 3.3.1.2 Handles.- The handles shall be fabricated of the material specified in 3.2.2.

* 3.4 Finish.-

* 3.4.1 Needles.- The surface of the needle shall be finished in accordance with the manufacturer's current commercial practice. The handles shall be treated for water resistance in accordance with the manufacturer's commercial practice.

* 3.5 Marking for identification.- The etching needle shall be marked on the handle with the manufacturer's name or trademark, and the style and size.

* 3.6 Workmanship.- The surfaces of the etching needles shall be free from slivers rough or sharp edges. The water resisting coating shall be a continuous, smooth, dry film, free from imbedded foreign matter and area of no film. Components shall not be broken, cracked, split, or malformed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves

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the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.- When a preproduction sample is required, it shall be examined for defects listed in table I and dimensions specified. The presence of any visual defect, any dimension not within the specified requirements, or failure to pass the test shall be cause for rejection of the preproduction sample.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.1.1 Intermediate test.- Prior to assembly, the needles shall be tested in accordance with 4.4.1 to determine compliance with 3.3.1.1. The lot shall be all the points of the same style and size offered for inspection at one time. The sample unit shall be one needle. The inspection level shall be II with an acceptable quality level (AQL) of 1.5, expressed in terms of defects per hundred units.

4.3.2 End-item inspection.- The lot shall be all the etching needles of one style and one size offered for inspection at one time. The sample unit shall be one etching needle.

4.3.2.1 Visual examination.- The end item shall be examined for the defects in table I. The inspection level shall be II and the AQL shall be 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not as specified	X	
	Handle not treated for water resistance	X	
	Not continuous, or is not smooth and uniform		X
	Area of no film		X
	Evidence of imbedded foreign matter		X

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TABLE I.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship	Wrong style or size	X	
	Any component broken, cracked, split or malformed	X	
	Slivers, rough or sharp edges on handle	X	
	Needle not centered, set and secured in handle	X	
Marking	Missing, incomplete, not legible		X

4.3.2.2 Dimensional examination.- Examination shall be made of the etching needles for defects in dimensions. Any dimension that is not within the specified requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.3.3 Examination of preparation for delivery.- Examination of preparation for delivery shall be in accordance with MIL-H-15424.

4.4 Test.-

4.4.1 Hardness.- Hardness tests to determine compliance with 3.3.1.1 shall be performed in accordance with ASTM E 18-65. Three readings shall be made on a needle of one style and one size. A reading shall be taken at each end of the needle and at the center of the needle. Any reading not within specified requirements shall constitute a defect.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.- Preservation, packaging, packing and marking for level A, B, or C shall be in accordance with MIL-H-15424 (see 6.2).

6. NOTES

* 6.1 Intended use.- The etching needles covered by this specification are to be used for hand etching metal plates in photoengraving, electro typing, and offset lithography.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Style and size required (see 1.2).
(c) First article.- When preproduction sample is required (see 3.1).
(d) Selection of applicable levels of preservation, packaging, and packing (see 5.1).

* 6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

* 6.4 Deletion data.- Style I, size 3, and style III, size 2 have been deleted from this specification, since they are no longer required.

Custodians:

Army - GL
Air Force - 84

Review activities:

Air Force - 84

User activities:

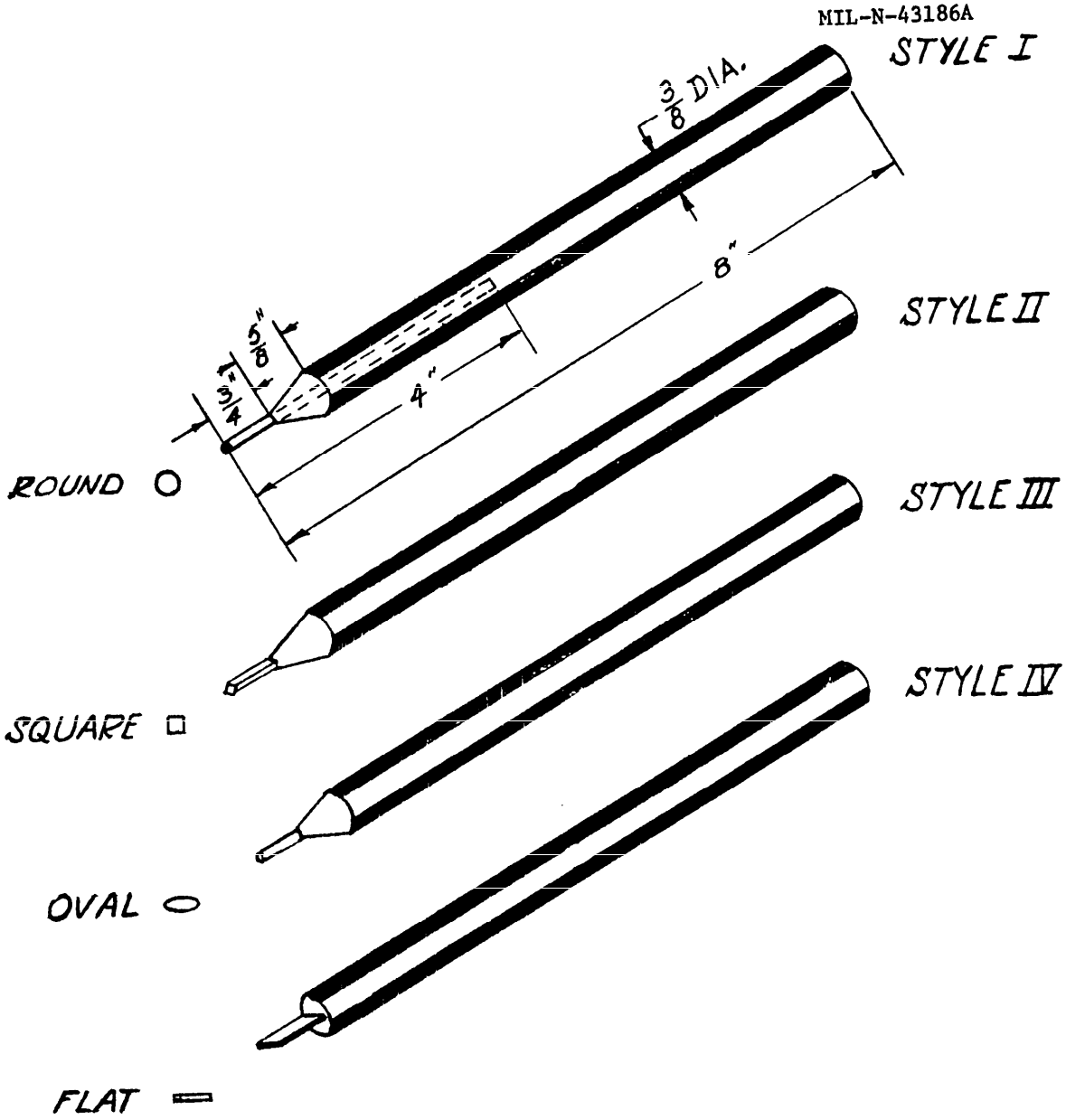
Army - ME
Navy - MC

Preparing activity:

Army - GL

Project No. 5120-0568

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DIMENSIONAL DATA (IN INCHES)				
SIZE	STYLE I ROUND	STYLE II SQUARE	STYLE III OVAL	STYLE IV FLAT
1	0.0312 DIA	0.027 SQ.	0.0625" BY 0.0312	0.0312 BY 0.1875
2	0.0625 DIA	0.035 SQ.	—	—
3	—	0.040 SQ.	—	—
4	0.1687 DIA	0.045 SQ.	—	—
5	—	0.055 SQ.	—	—
6	—	0.062 SQ.	—	—
7	—	0.075 SQ.	—	—

TOLERANCES: FRACTIONS $\pm \frac{1}{64}$ DECIMALS $\pm .002$

FIG 1. NEEDLES, ETCHING, HAND

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

NEEDLES, ETCHING, HAND, MIL-N-43186A

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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